

BRIEF REPORT

Galago Locomotion in Kibale National Park, Uganda

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Very few locomotor studies have been conducted on galagos. This is surprising given their interesting anatomy and ecology, as well as their increasing species diversity. In this study we investigated locomotion and postures in two sympatric galagos species (*Galagoides thomasi* and *Galago matschiei*) living in Kibale National Park, Uganda. *G. thomasi* uses arboreal quadrupedalism and leaping, while *G. matschiei* is more leaping-oriented. Both species utilize small oblique branches in the mid-canopy. These similarities in substrate use are most likely due to the similar body sizes and anatomies of the two species, as well as to the structure and availability of trees in Kibale National Park. Lastly, we compare the locomotor patterns of *G. thomasi* and *G. matschiei* with those observed in the few other quantitative locomotor studies available for galagos. Am. J. Primatol. 66:189–195, 2005. © 2005 Wiley-Liss, Inc.

Key words: *Galagoides thomasi*; *Galago matschiei*; positional behavior; Kibale National Park

INTRODUCTION

Galagos, or bushbabies, are a nocturnal group of arboreal prosimians that are distributed over most of sub-Saharan Africa [Anderson, 2000; Groves, 2001; Nash et al., 1989]. They occupy a variety of niches and are quite speciose [Bearder & Doyle, 1974; Bearder et al., 1995; Charles-Dominique, 1974, 1977; Honess, 1996; Groves, 2001; Nash et al., 1989]. Unfortunately, only a few species have been studied in detail. In this vein, only one wild quantitative study on galago locomotion has been published. Crompton's [1984] seminal work on *Galago moholi* and *Otolemur crassicaudatus* documented detailed locomotor and ecological differences between these two taxa. Other studies, such as those by Bearder and Doyle [1974], Charles-Dominique [1974, 1977], Crompton et al. [1987], Gebo [1987], Hall-Craggs [1965], Kingdon [1971], McArdle [1981], Napier and Walker [1967], Rowe [1996], and Walker [1974, 1979], discussed galago locomotion anecdotally or studied the animals in captivity. Additionally, we know

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of no anatomical study comparing the differences between these two taxa. Therefore, we thought that new data (especially quantitative data) regarding any of the many species of galagos, including *Galagoides thomasi* and *G. matschiei*, would benefit future anatomical and ecological studies in elucidating galago behavioral ecology and evolution. We addressed the following questions in this study: 1) What types of locomotor and postural behaviors are observed in *G. thomasi* and *G. matschiei*, and which occur most frequently? 2) What types of substrates are used most often? 3) How do Kibale Forest galago locomotor profiles compare with those of other small-bodied galagos?

MATERIALS AND METHODS

This study was conducted in Kibale National Park, which is located in western Uganda ($0^{\circ} 13' - 0^{\circ} 41' \text{ N}$ and $30^{\circ} 19' - 30^{\circ} 32' \text{ E}$) near the foothills of the Ruwenzori Mountains (Fig. 1). Skorupa [1988:44] described Kibale as a “mosaic of grassland, woodland thicket, colonizing forest, swamp forest, and high forest of several types.” However, he noted that there is no general consensus regarding Kibale vegetation. The park altitude ranges from 1,590 m in the north to 1,110 m in the south. During the summer months of data collection, the average minimum temperature recorded at the field station was 56.7°F , with an average maximum of 79.2°F . The average rainfall was 1.92 mm/day at the Kanyawara field station.



Fig. 1. Map showing location of Kibale National Park, Uganda.

We established two transects of 2.5 km and walked them nightly. To prevent systematic bias, we alternated the direction in which the transect was walked between samples of the same transect. We began the observations just before dusk at 1930 hr, using binoculars, a night scope, a headlight, and an additional flashlight. Species were identified either visually or by vocalizations (Bearder, personal communication). Each galago sighting provided a specific record of data (time of observation, initial height of the animal, positional behaviors used, and trees and substrates used). Individual animals were followed as long as possible, and all vocalizations were noted. A portable cassette recorder was used to record the data for transcription during daylight hours.

We followed Gebo and Chapman's [1995] protocol in recording positional behavior. Observations of single displacements were made with a new observation beginning whenever a change of position occurred. With each change of position, regardless of how long a bout lasted, a new record was started. Our locomotor and postural categories included quadrupedalism, leaping, vertical clinging and leaping, climbing, bipedal hopping, standing, sitting, and vertical clinging (see Gebo and Chapman [1995] and Hunt et al. [1996] for definitions of these categories). Likewise, we used three substrate size classes (large: >25 cm in circumference; medium and small: <5 cm) and three substrate angles (horizontal, <30°; oblique; and vertical, >60°) following Gebo and Chapman [1995]. We estimated vertical height use.

Numerous revisions of galago systematics have been published [Anderson, 2000; Bearder et al., 1995; Groves, 2001; Honess, 1996; Masters et al., 1994; Nash et al., 1989; Olson, 1979]. It is possible that Kibale Forest contains three galago species: *G. demidoff*, *G. matschiei*, and *G. thomasi*. However, *G. matschiei* and *G. thomasi* are the two species most often sighted (Bearder, personal communication) [Llorente et al., 2003; Weisenseel et al., 1993]. *G. thomasi* (55–149 g, mean=99 g [Nash et al., 1989] is larger than *G. demidoff*, and *G. thomasi* possesses a longer foot, skull, and ears [Masters & Bragg, 2000], and exhibits distinctive calls and penile morphology, ashy brown pelage, pale facial coloration, and a black strip down the dorsal surface of its tail [Bearder, 1999; Groves, 2001; Kingdon, 1997; Wickings et al., 1998]. *G. matschiei* is a medium-sized galago (196–225 g, mean=210 g [Nash et al., 1989]) and is distinguished by pointy nails [Hayman, 1937], a dark brown body color, eyes surrounded by black patches with a white line between them, and black-tipped ears [Bearder et al., 1995; Groves, 2001; Kingdon, 1997; Nash et al., 1989].

RESULTS

Table I shows the frequency of locomotor and postural behaviors for *G. thomasi* and *G. matschiei*. *G. thomasi* emphasizes arboreal quadrupedalism (35%) and leaping (23%) followed by bipedal hopping (15%) in its locomotor repertoire. For postures, vertical clinging represented half of the observations (54%). In comparison, *G. matschiei* is a more frequent leaper (31%) with far less use of arboreal quadrupedalism (13%). Bipedal hopping also occurs more frequently in *G. matschiei* (25%). Leaping, vertical clinging and leaping, and bipedal hopping account for 52% of the locomotor observations for *G. thomasi*, and an astounding 75% for *G. matschiei*. In contrast to the leaping and quadrupedal frequencies, climbing occurs at same frequency in both species. Like *G. thomasi*, *G. matschiei* prefers vertical clinging as its favored posture, and both taxa exhibit very similar postural frequencies for standing, sitting, and vertical clinging (Table I).

Both *G. thomasi* and *G. matschiei* exhibited locomotor and postural behaviors most often on small oblique supports (Table II). Large substrates were used only minimally. Both species showed a very similar use of the available substrates in

TABLE I. Locomotor and Postural Behavior for *Galagoides thomasi* and *Galago matschiei*

Category	<i>G. thomasi</i>	<i>G. matschiei</i>
	n %	n %
Locomotion		
Arboreal quadrupedalism	91 35%	47 13%
Climbing	35 13%	47 13%
VCL	37 14%	69 19%
Leaping	60 23%	113 31%
Bipedal hopping	40 15%	92 25%
Total	263	368
Postures		
Standing	11 30%	14 32%
Sitting	6 16%	5 11%
Vertical clinging	20 54%	25 57%
Total	37	44

TABLE II. Locomotor and Postural Use of Substrates for *G. thomasi* and *G. matschiei*

Substrate use	<i>G. thomasi</i>	<i>G. matschiei</i>
	n %	n %
Small supports	235 76%	392 74%
Medium supports	62 20%	134 25%
Large supports	11 4	1 <1%
Total	308	527
Horizontal supports	70 28%	121 22%
Oblique supports	120 49%	278 51%
Vertical supports	61 24%	144 27%
Total	251	543

TABLE III. Vertical Height Use of the Canopy in *G. thomasi* and *G. matschiei*

Canopy use	<i>G. thomasi</i>	<i>G. matschiei</i>
	n %	n %
Upper canopy	22 9%	64 13%
Mid-canopy	157 63%	259 54%
Lower canopy	71 28%	161 33%
Total	250	484

terms of branch size and angle. *G. thomasi* uses horizontal and large supports more frequently than *G. matschiei* (Table II).

Table III shows the results for vertical height use of the canopy by *G. thomasi* and *G. matschiei*. Both species favor the mid-canopy. Vertical clinging and vertical clinging/leaping were observed only slightly more often in the lower canopy than in the mid-canopy. This is not surprising, because small-diameter, vertical woody plants like *Mimulopsis* and *Brillantaisia* provide a low-canopy habitat that is ideal for vertical clinging and vertical clinging/leaping.

DISCUSSION

G. thomasi and *G. matschiei* are sympatric in Kibale Forest, are similar in body size and in their arboreal adaptations, and utilize similar trees and

tree structures. Sympatric primates avoid competition by employing different modes of locomotion and spending different amounts of time on different substrates [Fleagle et al., 1981]. Therefore, we might expect differences in substrate use and occupation of different canopy levels reflecting such niche separation. However, while the arboreal quadrupedalism-leaping gradient does reflect differences in locomotion, the postural frequencies reflect the use of similar supports, as does their use of substrate size and angle. *G. matschiei* is clearly a more frequent leaper and exhibits bipedal hopping more in its locomotor profile. This quadrupedalism-leaping gradient distinction was previously demonstrated in galagos [Charles-Dominique, 1974, 1977; Crompton, 1984; McArdle, 1981].

When we compare our new data on sympatric populations of *G. thomasi* and *G. matschiei* with other quantitative data in the literature, a few similarities can be noted. Captive data regarding *G. demidoff* (44–97 g, which is similar in size to *G. thomasi* (55–149 g) [Nash et al., 1989]) at the Duke Primate Center show a similar leaping frequency [Gebo, 1987]; however, quadrupedalism occurred less frequently and climbing was observed more frequently in that captive study (Table IV).

For *G. matschiei* and *G. moholi* (210 g and 206 g, respectively [Nash et al., 1989]), quantitative locomotor frequencies for arboreal quadrupedalism, total leaping, and climbing are within 7% of each other. *G. moholi* is also an accurate leaper that is known to use bipedal hopping [Hall-Craggs, 1974; Kingdon, 1971; Rowe, 1996], and we observed a high frequency of bipedal hopping by *G. matschiei* in this study. However, both species occupy very different habitats. *G. moholi* typically occupies *Acacia* woodland and thornveld, while *G. matschiei* is a tropical forest species. This suggests that body size and body adaptations may influence locomotor behavior to a greater extent than microhabitats.

Table V compares substrate use across four species of galagos. All four species show very similar frequencies for horizontal support use. Charles-Dominique's [1974] data revealed that *G. demidoff* uses vertical supports more often (48%) than the other species (27–31%). *G. matschiei* and *Euoticus elegantulus* use oblique branches more frequently (51%) than *G. thomasi* (43%) and *G. demidoff* (30%). If you add oblique and vertical support use frequencies across the four galago species in Table V, these values are very similar (74–78%). Two taxa (*G. thomasi* and *G. demidoff*) break up their microhabitat substrate use by alternating their use of preferred substrate and oblique or vertical supports.

Since small galagos have been described as using the “fine branch niche” [Martin, 1979], a high frequency of small support use (<5 cm in circumference) would be expected in these four species. The results for *G. thomasi*, *G. matschiei*, and *G. demidoff* certainly support this interpretation (74–89%, Table V), but

TABLE IV. Locomotor Differences Between *G. thomasi* and *G. demidoff* and between *G. matschiei* and *G. moholi*

	Quadrupedalism	Leaping	Climbing
<i>G. thomasi</i>	35%	37%	13%
<i>G. demidoff</i> ^a	25%	40%	24%
<i>G. matschiei</i>	13%	50%	13%
<i>G. moholi</i> ^b	20%	57%	17%

^aCaptive data from Gebo [1987].

^bWild data from Crompton [1983].

TABLE V. Comparison of Substrate Use Between *G. thomasi* and *G. demidoff* and *G. matschiei* and *E. elegantulus*

	Horizontal	Oblique	Vertical	Support < 5 cm
<i>G. thomasi</i>	23%	43%	31%	80%
<i>G. demidoff</i> ^a	22%	30%	48%	89%
<i>G. matschiei</i>	22%	51%	27%	74%
<i>E. elegantulus</i> ^a	22%	51%	27%	40% ^b

^aData from Charles-Dominique [1974].

^bData from Charles-Dominique [1977].

those for *E. elegantulus* do not (40%). *E. elegantulus*, one of the needle-clawed galagos, uses large supports quite frequently. *G. matschiei*, a medium-sized galago with pointy nails [Groves, 2001; Hayman, 1937; Nash et al., 1989], exhibits the same support use by angle as that shown by *E. elegantulus* (Table V), but these two species differ in their use of large-diameter supports. Pointy, keeled nails should reflect an adaptation to enable greater vertical support use [Cartmill, 1974]; however, *G. matschiei* is more similar to the other small-bodied galagos than it is to *E. elegantulus* in this regard.

CONCLUSIONS

We add a new quantitative study on galago locomotion in Kibale Forest to the few other such studies in the literature. *G. thomasi*, a 100 g galago, prefers arboreal quadrupedalism and leaping, while *G. matschiei*, a 200 g galago, prefers leaping. Both species use vertical clinging and leaping (14% and 19%, respectively). Bipedal hopping is observed more frequently in *G. matschiei*. Postures, canopy, and substrate use are very similar for these two galagos. Relative to other small- or medium-sized galagos, Kibale Forest galagos show a similar exploitation of their arboreal environment, but differ in their quadrupedalism-leaping locomotor gradient.

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